

test things like shell scripts, Chef cookbooks, Puppet modules, and more, using local virtualisation such as VirtualBox or VMware. Then, with the same configuration, you can test these scripts on remote clouds such as AWS or RackSpace with the same workflow. Ditch your custom scripts to recycle EC2 instances, stop juggling SSH prompts to various machines, and start using Vagrant to bring some sanity into your life.

For designers, Vagrant will automatically set up everything that is required for a particular Web app in order for you to focus on doing what you do best: design. Once Vagrant is configured, developers do not ever need to worry about how to get that app running. You will no longer need to bother other developers to help you fix your environment so that you can test your designs. Just check out the code, type ‘vagrant up’ and start designing.

With the implication that all designers could get others to do the configuring, I think you’ll agree that the “Just check out the code... and start designing” premise is very compelling.

You don’t need Vagrant to develop your Web applications on virtual machines. All you need is a virtualisation software package, something like VMware Workstation or VirtualBox, and some code. Download the half-gigabyte operating system image you want and install it. Then download and configure the stack you’ll be working with—let’s say it is Apache, MySQL or PHP. Next, install some libraries (CuRL and ImageMagick maybe) and, finally, configure the ability to easily copy files from your machine to the new virtual one, somewhat like Samba, or install an FTP server. Once all this is done, copy the code over, import the database, configure Apache’s virtual host, restart and keep your fingers crossed.

If you’re a bit weird like me, then the above tasks will be pretty easy to do and, secretly, quite a bit of fun.

Or you could use Vagrant. It allows you, or someone else, to specify in plain text how the machine’s virtual hardware should be configured and what should be installed on it. It also makes it insanely easy to get the code on the server. So check out your project, type ‘vagrant up’ and start work.

Installing Vagrant

Step 1 - Installing VirtualBox: First, download VirtualBox and install it. On *nix systems (Mac OSX, Linux, etc), you will need to modify your `bash_profile` (or `zsh_profile`) to extend your `$PATH` variable:

```
PATH=$PATH:/Applications/VirtualBox.app/Contents/MacOS/
export PATH
```

This will allow Vagrant to know where VirtualBox is installed, and this, of course, will vary for different operating systems. For Windows, installing VirtualBox is enough.

Step 2 - Installing Vagrant: To install Vagrant, just go to <https://www.vagrantup.com/downloads.html> and

download the installer, which will automatically add Vagrant to your system path so that it is available in the terminals. If it is not found, please try logging out and logging back into your system (this is particularly necessary sometimes for Windows).

Configuring Vagrant

Vagrant will create a custom virtual machine. To do this, Vagrant will need a template, or base VM, to copy and customise. The Vagrant development team is nice enough to provide a number of base Linux VMs. To install one, run the following command:

```
$ vagrant box add precise32 http://files.vagrantup.com/
precise32.box
```

This will download a base VM of Ubuntu 12.04 32-bit, which is about 230MB in size and will be referred to by the name ‘precise32’. Next, you’ll need a project to work on, and a development environment configuration. Here’s a Node.js demo; clone this repository, and move into the project directory, as follows:

```
$ git clone https://github.com/Srijancse/Vagrant-Node
$ cd Vagrant-Node
```

Take a look at the directory:

```
vagrant-node-mongo /
  README.md
  Vagrantfile
  /app
  /cookbooks
```

Inside the ‘app’ sub-directory, there is a simple `node.js` app that returns some text output to an HTTP request. In ‘cookbooks’ there are some Chef cookbooks, or collections of recipes that define automated configuration commands to install Node.js, MongoDB, apt-get, and other essential tools. Let’s take a closer look at the file named *Vagrantfile*. This is the Vagrant configuration file, which defines how to set up your VM project. It should look like what’s shown below:

```
Vagrant::Config.run do |config|
  config.vm.box = "precise32"
  config.vm.forward_port 3000, 3000
  config.vm.share_folder "app", "/home/vagrant/app", "app"
  # allow for symlinks in the app folder
  config.vm.customize ["setextradata", :id, "VBoxInternal2/
  SharedFoldersEnableSymlinksCreate/app", "1"]
  config.vm.customize ["modifyvm", :id, "--memory", 512]
  config.vm.provision :chef_solo do |chef|
    chef.cookbooks_path = "cookbooks"
    chef.add_recipe "apt"
```